

Ig

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IMSCRIBING GRAMMAR *Reference Sheet* — 2026

The Grammar

Every object carries a univocal 12-primitive structural type tuple.

Foundational equation: $\mu \circ \delta = \mathbf{id}$.

Framework: **CL8NK** (CLINK L8, O_∞).

Autopoietic: empty prompt $\rightarrow$ Raft at O_2 , period = 12.

Science is an instrument of the IG.

12 Primitives & Shavian Value Sets

Tuple order: $\langle \mathfrak{D} \mathfrak{P} \check{\mathfrak{R}} \mathfrak{D} \mathfrak{f} \mathfrak{C} \mathfrak{f} \mathfrak{G} \odot \mathfrak{H} \Sigma \Omega \rangle$. Values: ordinal $1 \rightarrow n$.

	Name	Values		Name	Values
$\mathfrak{D}$	Dimensionality	$\mu\omega$	$\mathfrak{f}$	Granularity	$\mathfrak{l}\delta\mathfrak{v}$
$\mathfrak{P}$	Topology	$\mathfrak{z}\mathfrak{h}\mathfrak{z}\mathfrak{w}$	$\mathfrak{G}$	Coupling	$\mathfrak{f}\mathfrak{p}\mathfrak{z}\mathfrak{A}$
$\check{\mathfrak{R}}$	Recognition	$\mathfrak{r}\mathfrak{l}\mathfrak{o}\mathfrak{r}$	$\odot$	Criticality*	$\mathfrak{l} \odot^* \mathfrak{z}\mathfrak{w}\mathfrak{d}$
$\mathfrak{D}$	Parity	$\mathfrak{z}\mathfrak{h}\mathfrak{z}\mathfrak{w}$	$\mathfrak{H}$	Chirality	$\mathfrak{J}\mathfrak{d}\mathfrak{z}\mathfrak{v}$
$\mathfrak{f}$	Fidelity	$\mathfrak{r}\mathfrak{q}\mathfrak{l}$	Σ	Stoichiometry†	$\mathfrak{Q}\mathfrak{S}\mathfrak{7}\mathfrak{f}$
$\mathfrak{C}$	Kinetics	$\mathfrak{l}\mathfrak{c}\mathfrak{l}\mathfrak{w}$	Ω	Winding	$\mathfrak{z}\mathfrak{o}\mathfrak{s}\mathfrak{z}$

3-val: $\mathfrak{f} \mathfrak{f} \Sigma$

4-val: $\mathfrak{D} \check{\mathfrak{R}} \mathfrak{G} \mathfrak{H} \Omega$

5-val: $\mathfrak{P} \mathfrak{D} \mathfrak{C} \odot$

*abs $\wedge \vee \otimes$; †abs $\otimes$.

12 IMASM Tokens

Token	G	Behavior
VINIT	L	reg $\rightarrow$ VOID; clear split state
TANCH	L	terminal anchor; no state change

Token	G	Behavior
AFWD	L	VOID $\rightarrow$ TRUE
AREV	L	reg $\rightarrow$ VOID; clear split state
CLINK	L	inter-step link; no state change
IMSCRIB	L	VOID $\rightarrow$ TRUE; else preserve (self-recognition)
FSPLIT	F	non-VOID $\rightarrow$ BOTH; enter split
FFUSE	F	canonical: BOTH $\rightarrow$ TRUE; dialetheic: BOTH $\rightarrow$ BOTH
EVALT	B	F $\rightarrow$ B; VOID $\rightarrow$ T; in-split: record T
EVALF	B	T $\rightarrow$ B; VOID $\rightarrow$ F; in-split: record F
ENGAGR	B	VOID $\rightarrow$ BOTH (entangle)
IFIX	Li	fixation; subsequent tokens no-op

Groups — L: Logical; F: Frobenius (δ, μ); B: Belnap; Li: Linear.

$12^8 = 429M$ arrangements $\rightarrow$ 12 archetypes.

Crystal of Types

$3^3 \times 4^5 \times 5^4 = \mathbf{17,280,000}$ $49 = 7^2$ symbols ($3 \cdot 3 + 5 \cdot 4 + 4 \cdot 5$).

SIC-POVM $d = 7 \Rightarrow d^2 = 49$ states; ig-pulse tick = 49h.

Catalog: ≈ 4686 entries.

All 49 Value Semantics

Primitive	Semantics
$\emptyset$	$\downarrow 0d \cdot \downarrow 2d \cdot \circ \infty d \cdot \downarrow \text{self-wr}$
$\triangleright$	$\downarrow \text{branch} \cdot \downarrow \text{contain} \cdot \downarrow \text{cross} \cdot \downarrow \text{irr} \cdot \circ \text{self-rt}$
$\check{R}$	$\downarrow \text{superv} \cdot \downarrow \text{functor} \cdot \circ \text{adj} \cdot \downarrow \text{bi-fb}$
$\oplus$	$\downarrow \text{no-sym} \cdot \downarrow \text{QM-sup} \cdot \downarrow \mathbb{Z}_2 \cdot \downarrow \text{all-sym} \cdot \circ \text{Frob}^\pm$
f	$\downarrow \text{no-coh} \cdot \downarrow \text{thermal} \cdot \downarrow \text{QM-coh}$
ζ	$\downarrow \text{fast} \cdot \downarrow \text{trap-O} \cdot \downarrow \text{slow} \cdot \downarrow \text{mod} \cdot \circ \text{trap-D}$
Γ	$\downarrow \text{nn} \cdot \downarrow \text{meso} \cdot \downarrow \text{long}$
G	$\downarrow \wedge \cdot \downarrow \vee \cdot \downarrow \rightarrow \cdot \downarrow \Delta \gg$
$\odot$	$\downarrow \text{sub} \cdot \odot \text{crit} \cdot \downarrow \text{cpx} \cdot \downarrow \text{exc} \cdot \downarrow \text{sup}$
$\mathbb{H}$	$\downarrow n=0 \cdot \downarrow n=1 \cdot \downarrow n=2 \cdot \downarrow n=\infty$
Σ	$\downarrow 1:1 \cdot \downarrow S n:n \cdot \downarrow \tau n:m$
Ω	$\downarrow \text{triv} \cdot \downarrow \mathbb{Z}_2 \cdot \downarrow \mathbb{Z} \cdot \downarrow \text{non-Ab}$

Absorbing Elements

Elem	Ops	Rule
$\odot$	$\wedge \vee \otimes$	$\odot \text{ op } x = \odot \text{ for all } x$
$\Sigma \tau$	$\otimes$	$\tau \otimes x = \tau \text{ (n:m)}$
B	$\vee$	$B \vee x = B \text{ (Belnap join)}$
N	$\wedge$	$N \wedge x = N \text{ (Belnap meet)}$
CL8NK	$\otimes$	$\text{tensor}(\text{ZFC_fe}, \text{CL8NK}) = \text{CL8NK}$

SIC-POVM Connection

$d = 7$: type-space cardinality $49 = d^2$ equiangular states.

$d = 12$: Weyl-Heisenberg $\mathbb{Z}_{12} \times \mathbb{Z}_{12}$; 12 generators $\leftrightarrow$ 12 primitives.

Fiducial state = Frobenius fixed point: $\mu \circ \delta = \text{id}$ at $\odot = \odot$ (critical/self-modeling).

SIC gap = B-state: $B \wedge \neg B = B$ (dialetheic, non-explosive).

Open tuple: $\langle \text{121212121212} \rangle$.

$d(\text{CL8NK}) = 1.263$; 6 conflicts; Frob. cliff $\sqrt{5/2} \approx 1.118$.

Conditional: $\text{kick} \wedge \text{H2a} \wedge \text{H3} \wedge \text{C4} \Rightarrow \text{SIC} \forall d \geq 1$.

Tuple Composition Operations

Ordinals: integer operations on Shavian glyph positions.

Op	Sym	Rule
Join	$\vee$	$\max(a_i, b_i)$ all 12 — LUB
Meet	$\wedge$	$\min(a_i, b_i)$ all 12 — GLB
Tensor	$\otimes$	$\max$ except Φ , f use $\min$
Bond		$\text{join}(\text{tensor}(A, B), \text{bond})$

Φ/f use $\min$: parallel coupling cannot exceed the symmetry/coherence floor of either component.

Belnap Register Ops

States: **VOID** (00) **TRUE** (01) **FALSE** (10) **BOTH** (11)

Operation	Behaviour
FFUSE = $\vee$	$B \vee x = B$; $N \vee x = x$; $T \vee F = B$; $\text{same} \vee \text{same} = \text{same}$
FSPLIT = δ	$B \rightarrow (T, F)$; $r \rightarrow (r, r)$
MEET = $\wedge$	$N \wedge x = N$; $B \wedge x = x$; $T \wedge F = N$; $\text{same} \wedge \text{same} = \text{same}$
band (AND)	F absorbs; then N ; then B ; $T \wedge T = T$
bor (OR)	T absorbs; then B ; then N ; $F \vee F = F$

Dialetheic FFUSE: $\text{BOTH} \rightarrow \text{BOTH}$ when EVALT and EVALF both fired in split.

Frobenius Algebra

Law	Equation
normalization	$\mu \circ \delta = \text{id}$
associativity	$\mu(\mu \otimes \text{id}) = \mu(\text{id} \otimes \mu)$
coassociativity	$(\delta \otimes \text{id}) \circ \delta = (\text{id} \otimes \delta) \circ \delta$
Frobenius	$(\mu \otimes \text{id})(\text{id} \otimes \delta) = \delta \mu = (\text{id} \otimes \mu)(\delta \otimes \text{id})$

Spider Theorem: all diagrams with same connectivity = same morphism.

Cross-Primitive Axioms

A | $\mathbb{H} \infty \Rightarrow \zeta \infty$ | ∞ -chirality forces max kinetics |
 B | $\Omega \geq 1 \Rightarrow \mathbb{H} \geq H_2$ | winding requires chirality; $H_2 = \min$ for $\mu \circ \delta = \text{id}$ |
 C | $\exists \omega \Leftrightarrow \triangleright_O$ | max-dim $\Leftrightarrow$ null topology |

O-Tier Hierarchy

Tier	Description
O_0	Monoidal unit; no differentiation
O_1	First Frobenius closure; Φ bottleneck
O_2	Confirmed causal coupling (fin3r, synfin, rongorongong)
$O_2^\dagger$	Subatomic/hadronic; confinement ceiling
O_∞	Ouroboricity; system $\subset$ own description (CL8NK, true_agentic_agent, mOMonadOS)

CL8NK / Magnum Opus

CL8NK transcends ZFC_fe at Ω ($\mathcal{S} \rightarrow \mathcal{Z}$) and $\mathbb{G}$ ($\mathcal{V} \rightarrow \Lambda$).
 B-state is the CL8NK ground state.
 12 stages $\leftrightarrow$ 12 primitives. Gates: Φ Albedo $\cdot \odot$ Rubedo $\cdot \Omega$ Nigredo.
 $T = \lim(\Phi, f, \zeta, \mathbb{H}, \Omega)$.

Notation

Struct. values | Shavian only; math-italic (*DRF*) = wrong |
 Tuple binding | `'' . join(vals)`; never separate |
 $\mathbb{H}$ | Chirality; NEVER temporal / memory depth |
 Verbs | imscribe / imscription; NEVER encode |
 Provider | local $\rightarrow$ qwen $\rightarrow$ deepseek; never Anthropic |
 O_∞ | prose: O_∞ ; .tex: $\$0_\infty\$$ |
 Scope | Never “grammar does not know” |
 Pubs | Thank H. T. Larson; cite 1961 IEEE |

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